

EDUGENIE – PHASE 6: PROJECT TESTING

Test Cases – AI-Powered Learning Assistant

Test Case ID	Module	Test Scenario	Test Steps	Test Data / Input	Expected Result	Actual Result	Status	Remarks
TC-001	Application	Launch EduGenie application	1. Start the FastAPI server 2. Open the application URL	http://127.0.0.1:8000	EduGenie home page loads successfully.	EduGenie home page loaded successfully at http://127.0.0.1:8000.	Pass	Application launched successfully.
TC-002	Q&A	Ask a basic question	1. Open Q&A feature 2. Enter a question 3. Submit	What is artificial intelligence?	A relevant AI-generated answer is displayed.	The system generated a relevant explanation of artificial intelligence.	Pass	Response was relevant to the question.
TC-003	Q&A	Ask a subject-related question	1. Open Q&A 2. Enter a study question 3. Submit	Explain photosynthesis.	The system returns a clear educational explanation.	The system returned a clear educational explanation of photosynthesis.	Pass	Explanation was understandable and educational.
TC-004	Explanation	Generate an explanation for a topic	1. Open Explanation module 2. Enter a topic 3. Submit	Explain machine learning in simple terms.	The system provides a simple and understandable explanation.	A simple explanation of machine learning was generated successfully.	Pass	Response was written in simple language.
TC-005	Summary	Generate a summary	1. Open Summary module 2. Enter text/topic 3. Submit	Artificial intelligence is used in many areas such as education and healthcare.	A concise summary is generated.	A concise summary was generated from the provided text.	Pass	Summary was shorter than the original input.
TC-006	Quiz	Generate a quiz	1. Open Quiz module 2. Enter a topic 3. Submit	Computer science basics	A relevant quiz is generated for the selected topic.	A topic-based quiz was generated successfully.	Pass	Quiz content matched the requested topic.
TC-007	Learning Path	Generate a learning path	1. Open Learning Path module 2. Enter a learning goal 3. Submit	Learn Python programming	The system generates an ordered learning path.	An ordered learning path for learning Python programming was generated.	Pass	Learning steps were presented in a logical order.
TC-008	Validation	Submit empty input	1. Open a learning module 2. Leave input blank 3. Submit	Blank input	The system prevents invalid submission or displays a suitable validation message.	The application displayed a validation message and did not process the blank input.	Pass	Invalid blank input was handled correctly.
TC-009	API	Verify API endpoint response	1. Open API documentation 2. Execute a supported endpoint 3. Check response	Valid API request	The API returns a successful response with the expected data.	The API returned a successful response containing the generated result.	Pass	Endpoint responded successfully.
TC-010	Error Handling	Handle invalid or unavailable AI request	1. Submit an invalid/problematic request 2. Observe application response	Invalid request / temporary API failure	The application shows a meaningful error message without crashing.	The application displayed an error message without crashing when the request could not be processed.	Pass	Error was handled without application crash.
TC-011	UI	Check navigation between modules	1. Open the home page 2. Navigate through available modules	Q&A, Explanation, Summary, Quiz, Learning Path	Navigation works correctly and the selected module opens.	Navigation between the available learning modules worked correctly.	Pass	Module navigation worked as expected.
TC-012	Integration	Verify frontend and backend integration	1. Enter valid input in the UI 2. Submit 3. Observe returned result	Valid educational question	Frontend successfully communicates with backend and displays the AI result.	The frontend successfully sent the request to the backend and displayed the returned AI result.	Pass	Frontend-backend communication worked successfully.

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EduGenie – Phase 6 Testing Guide	
Status values	
Not Executed	Test has not yet been performed.
Pass	Actual result matches the expected result.
Fail	Actual result does not match the expected result.
Usage	
Fill in Actual Result, Status, and Remarks after executing each test case.	
Documentation note	
The completed test records in the main sheet are sample project records. For a strictly evidence-based submission, replace the Actual Result and Status values after running the application.	